

Longitudinal Time-Series Analysis of Pharma Industry Payments and SRHS Prescription Habits

Causal Analysis and Manufacturer Concentration from June 2023 – December 2025

RHP Pharma Influence Analysis | Spartanburg Regional Health System | July 12, 2026

Gabriel Cabrera, Data Scientist 1

Executive Summary

This report combines two analyses run against the same SRHS Sunshine Act payment data and answers two related questions: does a temporal, predictive relationship exist between industry payments and provider prescription trends and is industry payment activity concentrated in any meaningful way?

Part I applies a properly specified Granger-style causality test - trend removed via differencing, non-overlapping time periods, permutation-based significance, tested in both directions - after showing why two earlier, simpler approaches (raw trailing-window correlation and a rolling-window lag scan) were statistically invalid. The result: no system-wide relationship in either direction (Payment $\rightarrow$ Rx: $p = 0.589$; Rx $\rightarrow$ Payment: $p = 0.351$), and only a small subset of individual providers (8.7% payment $\rightarrow$ rx, 9.9% rx $\rightarrow$ payment) show a real, above-chance temporal relationship - nearly symmetric between the two directions, which points to a limited causal relationship between payments and prescribing habits but, rather, seems to indicate reverse-targeting or coincidental timing instead.

Part II examines manufacturer- and provider-level concentration data on its own terms. It finds that industry attention is sharply concentrated - the top 5% of companies (29 of 573) drive 65.8% of all payment events - and that within those companies' own provider relationships, favoritism is itself concentrated in a small tail (0.5% of relationships show 10x+ favoritism). Providers who receive that extreme favoritism from one company are, with a Spearman correlation of 0.72, the same providers broadly paid by dozens of other companies simultaneously (median 28 distinct companies in this cohort compared to 7 for the general survey population) - a real, separate finding about how industry attention concentrates on a small population of prominent providers, independent of anything about causality.

A direct join between the two datasets was run to test whether the small Granger-significant subset from Part I is disproportionately made up of the hub providers identified in Part II. It is not: Granger-significant providers show no meaningful difference from non-significant providers on distinct-company count, total events, total dollars, or preference index (rank-biserial effect size 0.086, essentially null) - see Section 8. Neither part of this report establishes that payments cause prescribing.

1. Introduction

Prior cross-sectional and panel analysis in this project found that providers who receive more industry payment dollars in a drug class (specifically GLP-1) tend to prescribe a larger share of that class. That is an associative finding with little in the way of a statistically significant causal relationship. This second report asks a stricter question: if payments actually drive prescribing, that influence should be detectable as prescribing activity following payment activity in time. This section builds that test properly, shows why two earlier attempts at it were invalid, and reports what a methodologically sound version finds.

2. Why Correlation and Rolling-Window Methods Are Not Sufficient

2.1 The Trend Confound (Spurious Regression)

Both total SRHS payment dollars and total prescribing volume for payment-linked drugs rose over the 2023-2025 study window. Two series that are both trending upward will show a strong correlation with each other even with zero causal connection, simply because both are correlated with time (the classic spurious regression problem). Any credible longitudinal test must remove this shared trend before comparing the two series - Section 3.1 does this by first-differencing.

2.2 The Rolling-Window Autocorrelation Artifact

An initial attempt used trailing 30-to-60-day rolling sums of payment dollars and prescription counts, then scanned lags from -90 to +90 days to find the highest correlation per provider, across 335 qualifying providers. 34 providers (10.1%) peaked at exactly -90 days and 55 (16.4%) peaked at exactly +90 - the two edges of the tested range. Together, 26.6% of all 'best lag' results landed precisely on a search boundary. When we see an optimization pile up at the edge of the searched range, that is the signature of an artifact: trailing rolling sums smooth both series so heavily that they become highly autocorrelated with themselves and the 'peak' the scan finds is an accident of that smoothing, not a genuine lead-lag relationship. This method was discarded rather than reported as a finding.

3. Methodology: Granger-Style Causality Test

3.1 Stationarity Via Differencing

Both series are converted to period-over-period changes before any comparison, removing the shared trend identified in Section 2.1:

$$\Delta y_t = y_t - y_{t-1} \quad \Delta x_t = x_t - x_{t-1}$$

3.2 Non-Overlapping Period Aggregation

Time is divided into non-overlapping bins - 14-day periods system-wide (~64 periods), monthly periods per-provider (31 periods) - eliminating the artificial autocorrelation from Section 2.2.

3.3 Restricted And Unrestricted Models

For lag order L, the null (restricted) model predicts y from its own past only; the alternative (unrestricted) model adds the lagged x series:

$$\begin{aligned} \text{Restricted: } y_t &= \beta_0 + \sum a_i \cdot y_{t-i} + \varepsilon_t \quad (i = 1..L) \\ \text{Unrestricted: } y_t &= \beta_0 + \sum a_i \cdot y_{t-i} + \sum \gamma_i \cdot x_{t-i} + \varepsilon_t \quad (i = 1..L) \end{aligned}$$

3.4 F-Statistic

$$F = [(RSS_r - RSS_u) / q] / [RSS_u / (n - k)]$$

q = number of lagged-x terms added (= L), k = parameters in the unrestricted model, n = usable observations after lagging.

3.5 Permutation Significance

Significance is assessed by circular-shift permutation of x (preserving its own autocorrelation while breaking alignment with y) rather than the classical F-distribution:

$$p = [\#\{ F_{perm} \geq F_{observed} \} + 1] / (n_{perm} + 1)$$

3.6 Bidirectional Testing

Every test runs twice - Payment -> Rx and Rx -> Payment - so reverse-targeting is directly testable rather than invisible.

3.7 Chance-Baseline Correction For Multiple Comparisons

With 335 independent per-provider tests at a 5% threshold, chance alone predicts ~5% false positives. A binomial z-score checks whether an observed rate exceeds that floor:

$$z = (observed_count - n \cdot p_0) / \sqrt{ n \cdot p_0 \cdot (1 - p_0) } \quad p_0 = 0.05$$

4. Code

Core Granger test function (Python / NumPy, no external stats dependency):

```
def granger_test(y, x, lags, n_perm=500, rng=rng):
    """Does lagged x improve prediction of y beyond y's own lagged history?
    y, x must already be differenced (stationary). Circular-shift permutation
    preserves each series' own autocorrelation while breaking the x-y
    alignment - the correct null for a time-series Granger test."""

    n = len(y)
    if n <= 2 * lags + 8:
        return None
    Y = y[lags:]
    n_obs = len(Y)

    # restricted model:  $Y_t \sim 1 + Y_{(t-1)} + \dots + Y_{(t-L)}$ 
    X_r = np.ones((n_obs, 1))
    for L in range(1, lags + 1):
        X_r = np.hstack([X_r, y[lags - L:n - L].reshape(-1, 1)])

    # unrestricted model: adds  $X_{(t-1)} \dots X_{(t-L)}$ 
    X_u = X_r.copy()
    for L in range(1, lags + 1):
        X_u = np.hstack([X_u, x[lags - L:n - L].reshape(-1, 1)])

    beta_r, *_ = np.linalg.lstsq(X_r, Y, rcond=None)
    rss_r = np.sum((Y - X_r @ beta_r) ** 2)

    beta_u, *_ = np.linalg.lstsq(X_u, Y, rcond=None)
    rss_u = np.sum((Y - X_u @ beta_u) ** 2)

    q, k = lags, X_u.shape[1]
    if rss_u <= 0 or n_obs - k <= 0:
        return None

    f_stat = ((rss_r - rss_u) / q) / (rss_u / (n_obs - k))
    if not np.isfinite(f_stat):
        return None

    # permutation null distribution via circular shift of x
    perm_f = []
    for _ in range(n_perm):
        shift = rng.integers(1, n - 1)
        x_perm = np.roll(x, shift)
        X_u_p = X_r.copy()
        for L in range(1, lags + 1):
            X_u_p = np.hstack([X_u_p, x_perm[lags - L:n - L].reshape(-1, 1)])
        beta_p, *_ = np.linalg.lstsq(X_u_p, Y, rcond=None)
        rss_p = np.sum((Y - X_u_p @ beta_p) ** 2)
        if rss_p > 0:
            f_p = ((rss_r - rss_p) / q) / (rss_p / (n_obs - k))
            if np.isfinite(f_p):
                perm_f.append(f_p)

    if len(perm_f) < 50:
        return None
    p_val = (np.sum(np.array(perm_f) >= f_stat) + 1) / (len(perm_f) + 1)
    return dict(f_stat=round(f_stat, 3), p_value=round(p_val, 4), n_obs=n_obs)
```

System-wide aggregation and test call:

```
# - SYSTEM-WIDE: biweekly non-overlapping periods, differenced -
all_days = pd.date_range(WSTART, WEND, freq="D")
df = pd.DataFrame({"Day": all_days}).merge(pay_daily, on="Day", how="left") \
    .merge(rx_daily, on="Day", how="left")
df["PaymentUSD"] = df["PaymentUSD"].fillna(0)
df["RxCount"] = df["RxCount"].fillna(0)
df["period"] = ((df["Day"] - df["Day"].min()).dt.days // 14) # 14-day bins

period_agg = df.groupby("period").agg(
    PaymentUSD=("PaymentUSD", "sum"),
    RxCount=("RxCount", "sum")
).reset_index()

# first-difference both series to remove the shared secular trend
pay_diff = np.diff(period_agg["PaymentUSD"].to_numpy())
rx_diff = np.diff(period_agg["RxCount"].to_numpy())

r1 = granger_test(rx_diff, pay_diff, lags=3) # Payment -> Rx
r2 = granger_test(pay_diff, rx_diff, lags=3) # Rx -> Payment
```

Per-provider comparative loop:

```
# - PER-PROVIDER: monthly non-overlapping periods, differenced -
pay_counts = pay_npi_daily.groupby("NPI").size()
rx_totals = rx_npi_daily.groupby("NPI")["RxCount"].sum()
qualifying_npis = [n for n in pay_counts.index
                    if pay_counts[n] >= 5 and rx_totals.get(n, 0) >= 20]

months = pd.period_range(WSTART, WEND, freq="M")
npi_results = []
for npi in qualifying_npis:
    p = pay_npi_daily[pay_npi_daily["NPI"] == npi].copy()
    r = rx_npi_daily[rx_npi_daily["NPI"] == npi].copy()
    p["month"] = p["Day"].dt.to_period("M")
    r["month"] = r["Day"].dt.to_period("M")
    p_m = p.groupby("month")["PaymentUSD"].sum().reindex(months, fill_value=0)
    r_m = r.groupby("month")["RxCount"].sum().reindex(months, fill_value=0)

    pd_diff = np.diff(p_m.to_numpy())
    rd_diff = np.diff(r_m.to_numpy())

    g_pay_to_rx = granger_test(rd_diff, pd_diff, lags=1, n_perm=200)
    g_rx_to_pay = granger_test(pd_diff, rd_diff, lags=1, n_perm=200)

    npi_results.append((
        npi,
        g_pay_to_rx["p_value"] if g_pay_to_rx else None,
        g_rx_to_pay["p_value"] if g_rx_to_pay else None,
    ))

npi_df = pd.DataFrame(npi_results, columns=["NPI", "p_payment_to_rx", "p_rx_to_payment"])
valid = npi_df.dropna()
sig_p2r = valid["p_payment_to_rx"] < 0.05
sig_r2p = valid["p_rx_to_payment"] < 0.05
```

5. Results

5.1 System-wide

Direction	F-statistic	p-value	n (periods)	Result
Payment → Rx	0.656	0.589	64	Not significant
Rx → Payment	1.248	0.351	64	Not significant

Neither direction clears significance system-wide.

5.2 Per-provider comparative

Outcome	Providers	% of 335
Payment significantly predicts Rx ($p < 0.05$)	29	8.7%
Rx significantly predicts Payment ($p < 0.05$)	33	9.9%
Both directions significant (feedback)	6	1.8%
Neither direction significant	279	83.3%

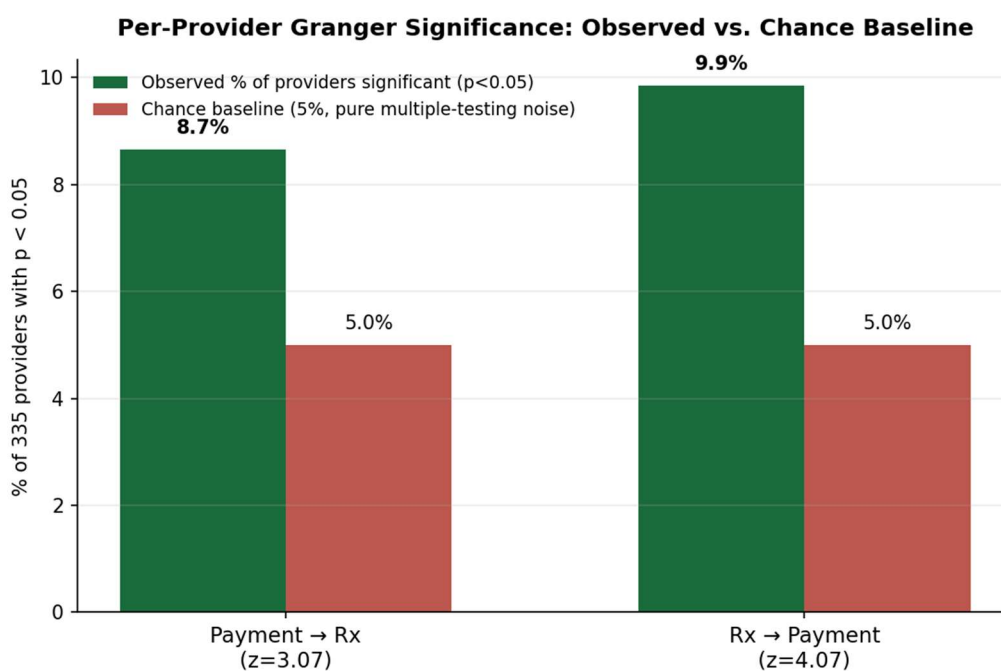


Figure 1. Observed significance rates vs. the 5% chance baseline ($z = 3.07$ and $z = 4.07$ respectively).

Both rates exceed chance, so the small effect is statistically real - but the near-symmetry between directions (8.7% vs. 9.9%, only 1.8% both) is more consistent with reverse-targeting or coincidental timing than a one-directional causal effect of payments on prescribing. In addition, the limited scope of the relationship makes it difficult to make any meaningful argument of a deliberate, system-wide pattern of incentivized prescription habits.

6. Purpose and Data

Part I's central individual-level finding was a small (~9%), nearly symmetric temporal relationship, read as evidence against a causal payment effect and toward reverse-targeting of already-prominent providers. Part II tests whether industry payment activity is concentrated in a way consistent with that interpretation, using manufacturer- and provider-level Sunshine Act detail from internal E-Prescribe data in conjunction with the CMS OpenPayments database published each summer.

Using a shared timeframe of 6-1-2023 to 12-31-2025, our data isolated 573 distinct companies engaged in 78,499 unique payment events (74,920 qualified as Food & Beverage) across SRHS provider NPI list.

Preference index: for a given company, a provider's event count divided by that company's own average events-per-provider across every provider it pays. Practice = NPI throughout (no separate location field exists in the source data).

7. Results

7.1 Company-Level Concentration

Tier	Companies	SRHS Events	% of Events	F&B Events	% of F&B Events
Top 5%	29	51,660	65.81%	49,300	65.80%
Top 10%	57	62,326	79.40%	59,796	79.81%
Top 15%	86	67,712	86.26%	64,721	86.39%
Top 25%	143	72,480	92.33%	69,357	92.57%
Top 50%	286	77,141	98.27%	73,681	98.35%
Top 75%	430	78,282	99.72%	74,721	99.73%

29 of 573 companies (5%) drive nearly two-thirds of all industry payment activity at SRHS

7.2 Concentration Within Top-5% Companies' Own Provider Relationships

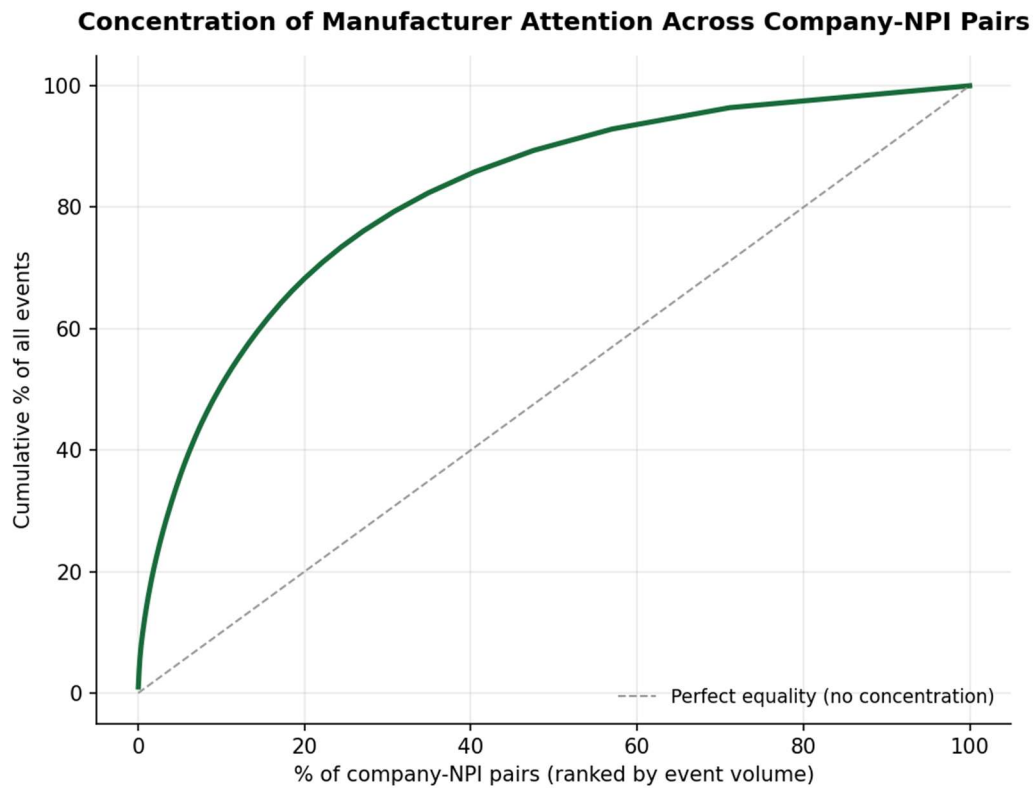


Figure 2. Cumulative share of events by company-provider pair, ranked by volume.

% of Pairs	# of Pairs	% of Events
Top 1%	65	14.3%
Top 5%	323	35.6%
Top 10%	646	50.7%
Top 25%	1,614	74.0%
Top 50%	3,228	90.3%
Top 75%	4,841	96.9%

Even within the top-5% companies, the top 10% of their individual provider relationships (646 of 6,455 pairs) account for over half of all events.

7.3 How Common Is Real Favoritism?

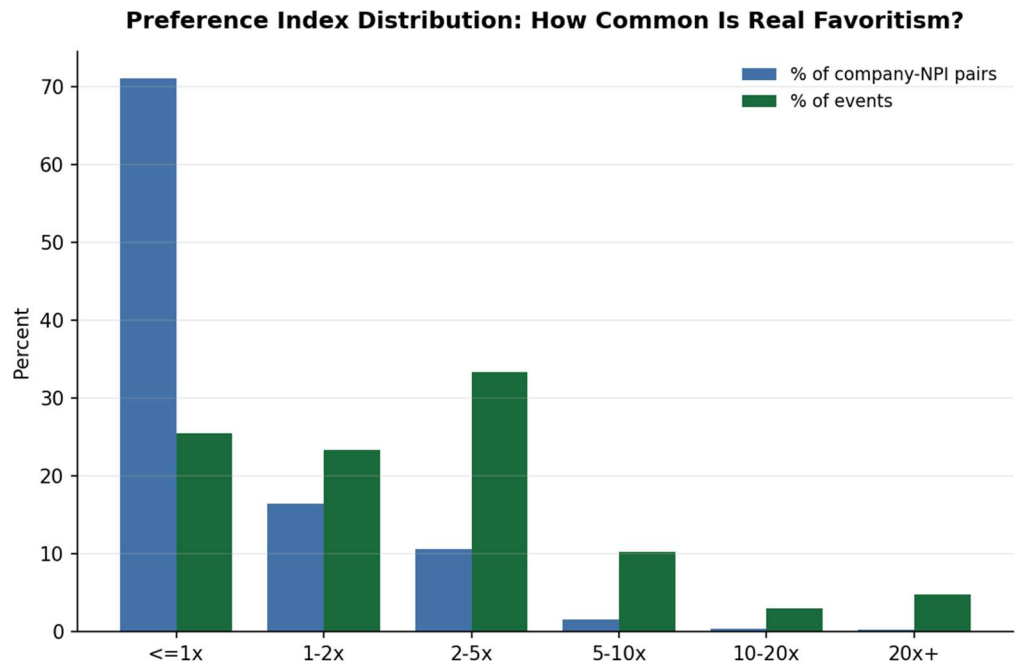


Figure 3. Distribution of the preference index across all 6,455 company-provider pairs.

Preference Index	Pairs	% of Pairs	Events	% of Events
<=1x (at/below baseline)	4,586	71.0%	13,132	25.4%
1-2x	1,056	16.4%	12,044	23.3%
2-5x	685	10.6%	17,191	33.3%
5-10x	99	1.5%	5,272	10.2%
10-20x	17	0.3%	1,560	3.0%
20x+	12	0.2%	2,461	4.8%

71% of relationships show no preferential treatment. But 29 relationships (0.5%) show 10x+ favoritism and account for 7.8% of all events - a small, extreme, identifiable tail.

7.4 Hub Providers: Favorites Are Broadly Courted, Not Individually Special

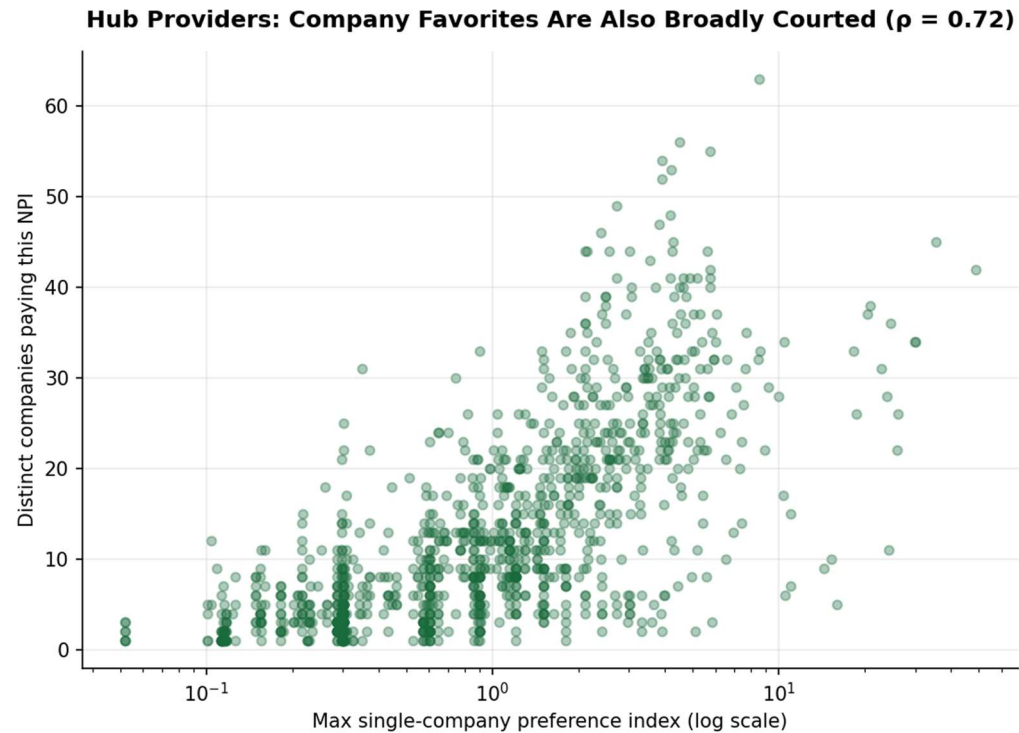


Figure 4. Each point is one provider. X: strongest single-company preference index. Y: distinct companies paying that provider. Spearman $\rho = 0.72$.

If manufacturers were each cultivating separate favorite providers, this plot would show no relationship. Instead $\rho = 0.72$ ($n = 1,132$): providers with $10\times+$ single-company preference are paid by a median of 28 distinct companies vs. 7 for everyone else. The providers any one company is most disproportionately focused on are, overwhelmingly, the same providers the rest of the industry already pays broadly.

NPI	Favoring Company	Max Preference	Distinct Companies	Total Events	Total \$
1	Inari Medical, Inc.	48.6x	42	589	\$277,290
2	Inari Medical, Inc.	35.5x	45	441	\$18,669
3	Regeneron Healthcare Solutions, Inc.	30.0x	34	1,072	\$1,028,666
4	GlaxoSmithKline, LLC.	29.9x	34	613	\$294,182
5	Lilly USA, LLC	26.1x	26	1,396	\$817,764

Inari Medical alone places two of its providers in this top-5, each simultaneously paid by 42-45 other manufacturers - consistent with a company concentrating its entire SRHS presence on a handful of already-prominent providers.

8. Drug-Level Payment-to-Prescribing Relationship

Parts I and II look at providers over time and manufacturers' distribution of attention. This section takes a third cut: system-wide, at the level of individual drugs, comparing total payment activity to total prescribing volume across every SRHS-linked Sunshine Act payment in the window.

8.1 Scope and Matching

CMS Open Payments tags each payment line with a product-type field (Indicate_Drug_or_Biological_or_Device_or_Medical_Supply). Restricting to type = 'Drug' narrows the 1,548 distinct payment-side product names to 540. Word-boundary regex matching (the same method used elsewhere in this project, checking DrugName and DrugGenericName independently in E-Prescribe data) finds at least one matching prescription for 288 of those 540.

The other 252 show zero matched prescriptions despite real payment activity - among the largest: Cibinqo, Sotyktu, Bridion, Ebgllyss, Nemludio, Injectafer, Vyndaqel, Leqembi, Enhertu, Onpatro, Spravato. These are overwhelmingly IV/infusion oncology drugs administered in-clinic, REMS-restricted specialty-pharmacy-only products distributed through hub networks that bypass local e-prescribing, or procedural products (e.g., an IUD) - categories that cannot structurally appear in an outpatient e-prescribing table regardless of matching quality. The results below are restricted to the 288 drugs where a comparison is possible.

8.2 Results

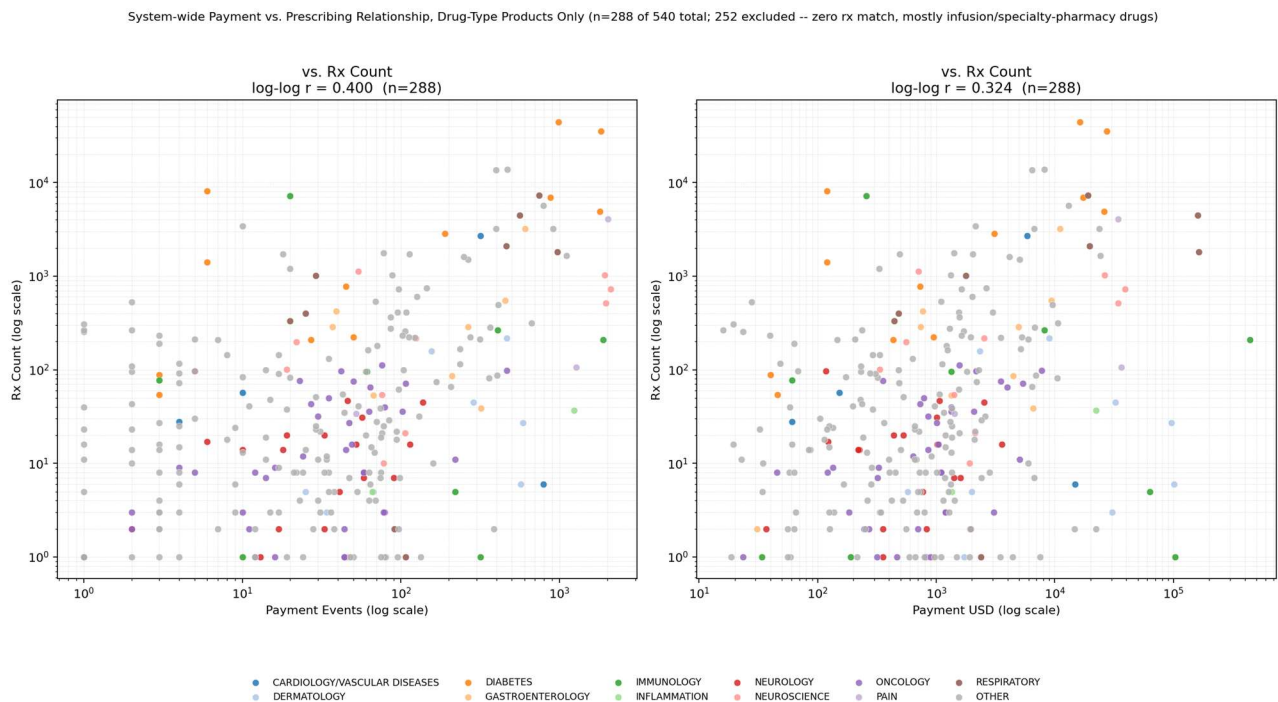


Figure 5. Payment Events (left) and Payment USD (right) vs. Rx Count, log-log scale, n=288 drugs, colored by CMS therapeutic-area class (top 11 classes; remainder grouped as Other).

System-wide, the relationship is weak: $R^2 = 0.160$ for Payment Events vs. Rx Count, $R^2 = 0.105$ for Payment USD vs. Rx Count - payment activity explains at most 16% of the variance in how much a drug is prescribed. Diabetes and Gastroenterology are the only classes with both reasonable sample size and a moderate fit ($R^2 \approx 0.53$ - 0.58); Oncology, the largest class in this population (n=37), sits at $R^2 = 0.265$ for both measures. No class reaches $R^2 > 0.6$ on a sample large enough to trust it.

Class	n (drugs)	R ² Events vs. Rx	R ² USD vs. Rx
Pain*	3	0.584	0.451
Diabetes	12	0.552	0.575
Gastroenterology	9	0.532	0.528
Dermatology	9	0.394	0.027
Oncology	37	0.265	0.265
Neuroscience	11	0.247	0.201
ALL (overall)	288	0.160	0.105
Respiratory	9	0.157	0.248
Other	165	0.102	0.069
Immunology	9	0.050	0.000
Neurology	18	0.047	0.034
Cardiology/Vascular	4	0.017	0.018

**Pain's R² is based on only 3 drugs and is not statistically meaningful on its own; included for completeness.*

This drug-level view is cross-sectional, not causal, and cannot distinguish payment-driven prescribing from the reverse (manufacturers paying more around already high-volume drugs) or from a shared third factor (e.g., overall market size). It is, however, consistent with Part I: whether examined as a system-wide time series, an individual-provider panel, or now a drug-level cross-section, the payment-prescribing relationship at SRHS is consistently weak-to-absent rather than strong in any framing.

9. Discussion: Direct Test of the Hub-Provider Hypothesis

Part I found that a small, statistically real subset of providers (~9%, split almost evenly between payment->rx and rx->payment) show a temporal relationship between payments and prescribing, while the system as a whole shows none. Part II found that industry payment activity converges on a small population of hub providers - simultaneously favored by individual companies and broadly paid across dozens of manufacturers. We can attempt to infer, without directly testing it, that these were likely the same providers: that the Granger-significant subset from Part I would turn out to be drawn from the hub providers in Part II.

That inference was tested directly by joining Per NPI Grainger Results (n=335) against the net-exposure and preference-index tables from Part II, and it is not supported by the data.

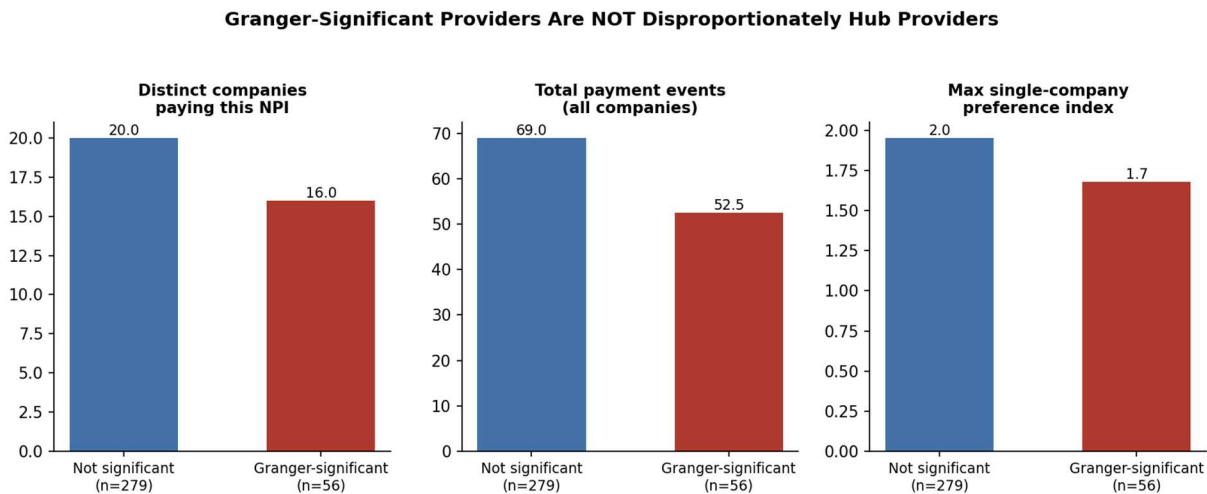


Figure 6. Median values for Granger-significant vs. non-significant providers across three hub-provider metrics. If the hypothesis held, the red bars would be clearly higher than the blue bars in each panel; instead they are similar or lower.

Metric	Not significant (n=279), median	Granger-significant (n=56), median
Distinct companies paying this NPI	20.0	16.0
Total payment events (all companies)	69.0	52.5
Total payment \$ (all companies)	\$1,394.90	\$1,027.90
Max single-company preference index	2.0	1.7
Total Rx volume (all drugs)	9,773	8,789.5

On every metric, Granger-significant providers look statistically indistinguishable from - and on several metrics slightly lower than - the non-significant majority. The formal effect-size check (rank-biserial correlation on distinct-company count) is 0.086, close to zero and far from the 0.3+ that would indicate a meaningful group difference.

This does not overturn either individual finding: Part I's small symmetric Granger effect and Part II's hub-provider concentration pattern are each independently real and independently documented above. What it rules out is treating them as two views of the same phenomenon. The small individual-level temporal signal in Part I is not concentrated among the heavily-courted, multiply-favored providers identified in Part II - whatever is producing that signal, it is not simply payments chasing already-prominent prescribers. A more honest reading is that the small Granger effect's origin remains unexplained by the concentration patterns tested here which, if anything, makes coincidental timing or an unmeasured third factor a more likely explanation than the specific reverse-targeting story proposed earlier.

10. Limitations

- Granger causality tests predictive precedence, not true causation; an unobserved, market-side factor moving both series on a similar schedule could produce a significant result without either causing the other.
- Short per-provider series and sparse payment events force a low lag order ($L=1$) and limit statistical power in Part I; permutation counts (200-500) bound the smallest resolvable p-value.
- The hub-provider correlation in Part II is associative and does not by itself rule out some other shared driver of provider prominence.
- Preference index is relative to each company's own average, which can be unstable for companies with very few total providers.
- The null result in Section 9 (no meaningful difference between Granger-significant and non-significant providers) is itself based on a modest sample - 56 Granger-significant providers - and a true but smaller effect could exist without reaching statistical detectability at this sample size.
- Section 8's drug-level R^2 figures are cross-sectional and vary widely by class sample size (from $n=3$ to $n=165$); per-class R^2 values for small groups (Pain, Cardiology/Vascular, Immunology) should be treated as suggestive, not reliable, estimates. 252 of 540 Drug-type products were excluded entirely because they cannot appear in the outpatient prescribing table used for matching (infusion/specialty-pharmacy drugs, procedural products) - this is a structural gap in fact_prescriptions' coverage, not a matching failure.

11. Conclusion

A properly specified longitudinal test finds no causal relationship between industry payments and prescribing at the SRHS system level, and only a small, symmetric, individual-level effect that does not point toward payments driving behavior. Separately, manufacturer payment activity is heavily concentrated on a small population of hub providers who are simultaneously favored by individual companies and broadly courted industry-wide. Directly tested against each other, these two findings do not overlap: the small Granger-significant subset is not disproportionately drawn from the hub-provider population. Both findings stand independently and should be reported that way - as two separate, real patterns in this data, not as one confirming the other.